Califnco — Round 1 Technical Assignment

Software Developer Hiring | Deadline: 25 October 2026 by 11:59 PM IST

1. Mobile App *(Mandatory — Android)*

Build an E-Commerce Water Purifier App using any cross-platform framework and Firebase (₹0 cost).

- **Customer:** Catalog of 50+ items (purifiers, dispensers, pumps, filters, spares). *AI/mock data is fine, but images must match product names and look clean/realistic.* Cart, checkout, order history, service booking, profile, push notifications.
- **Dealer:** Separate login (`role: "Dealer"`). Sees the same products at a **lower wholesale price**, places bulk orders, order history.
- **Admin Panel:** View/manage orders, approve/reject dealers, service desk. **Must be deployed live on Vercel.**

Suggested Free Tools:

- **Database & Auth:** Firebase Auth + Firestore (free Spark plan)
- **Push Alerts:** Firebase FCM (free)
- **Product Images:** Cloudinary or ImgBB *(Firebase Storage is not free)*
- **Payments:** Razorpay Test Mode (`rzp_test_` keys — zero KYC/documents needed)
- **Location:** OpenStreetMap, MapTiler, or Geoapify *(Google Maps requires credit card)*
- **Notifications:** EmailJS (email) · TextBee (SMS)

2. AI Automation *(Mandatory)*

Automate posting 1 product daily to Instagram with an **AI-generated image** and **AI-generated caption + hashtags** without manual work.

- **Suggested Stack:** n8n (Docker) or Make.com · Meta Graph API · Google Sheets · Groq or Gemini API (captions) · Hugging Face Inference API (image gen) · Cloudinary/ImgBB (public hosting).

> 💡 **Tool Flexibility & Free Tiers:** All tools suggested across both parts have free tiers based on our knowledge. However, if any tool has recently changed its free policy or is unavailable in your region, please find and use any other zero-cost tool to resolve it. What matters is achieving the exact same result—just state your alternative choice and reason in your submission.

⭐ Bonus (+15 pts)

- **WhatsApp Status (+10):** Post the daily product to WhatsApp Status via **WAHA** (free open-source, run locally via Docker). Use a secondary number.
- **App Polish (+5):** Animations, image lightbox/zoom, referral system, dark mode.

Scoring (100 pts)

Category	Evaluation Criteria	Points
Mobile App (75 pts)	Core app functionality (Customer, Dealer, Admin working)	30
	UI polish & realistic, relevant product imagery	15
	Code structure, cleanliness & architecture	10
	Firebase integration (Auth, Firestore, FCM alerts)	8
	Razorpay checkout flow (test mode)	5
	Maps / Address integration (free alternative)	4
	Email & SMS notifications working	3
AI Automation (25 pts)	Scheduled daily workflow (runs without manual input)	10
	AI-generated product image in post	8
	AI-generated caption with hashtags	7
Bonus (+15 pts)	WhatsApp Status automation (via WAHA)	+10
	In-app extras (animations, zoom, dark mode, referral)	+5

> ⚠️ **Quality over Quantity:** A few polished, working features will always score higher than an app trying to do everything with bugs or broken flows.

Submission

Please submit your solution through the official **Round 1 Submission Form** (also linked in your invitation email).

> ⚠️ **Google Drive Permissions:** Ensure all links submitted in the form are set to **"Anyone with the link can view"** — locked links will be skipped.

AI coding tools are allowed. We evaluate whether you understand what you built and whether it works.

All the best,
Al Amin
Califnco